

**DATA STRUCTURES AND ALGORITHMS
(CSEN 2101)**

Time Allotted : 3 hrs

Full Marks : 70

Figures out of the right margin indicate full marks.

*Candidates are required to answer Group A and
any 5 (five) from Group B to E, taking at least one from each group.*

Candidates are required to give answer in their own words as far as practicable.

**Group – A
(Multiple Choice Type Questions)**

1. Choose the correct alternative for the following: **10 × 1 = 10**

- (i) Consider a max heap, represented by the array. Now consider that a value 35 is inserted into this heap.

Array Index	1	2	3	4	5	6	7	8	9
Value	40	30	20	10	15	16	17	8	4

After insertion, the new heap is

- (a) 40, 30, 20, 10, 15, 16, 17, 8, 4, 35 (b) 40, 35, 20, 10, 30, 16, 17, 8, 4, 15
(c) 40, 30, 20, 10, 35, 16, 17, 8, 4, 15 (d) 40, 35, 20, 10, 15, 16, 17, 8, 4, 30

- (ii) What is the output of following function for start pointing to first node of following linked list?

1->2->3->4->5->6

void fun(struct node* start)

```
{  
    if(start == NULL)  
        return;  
    printf("%d ", start->data);  
    if(start->next != NULL )  
        fun(start->next->next);  
    printf("%d ", start->data);  
}
```

- (a) 1 4 6 6 4 1 (b) 1 3 5 1 3 5 (c) 1 2 3 5 (d) 1 3 5 5 3 1

- (iii) A stack S has the entries a,b,c with 'a' on the top. Another stack T is empty. An entry popped out of the Stack S can be printed immediately or pushed in to the stack T, finally popped out of the Stack T and printed. Then which sequence can never be printed

- (a) b a c (b) b c a (c) c a b (d) a b c

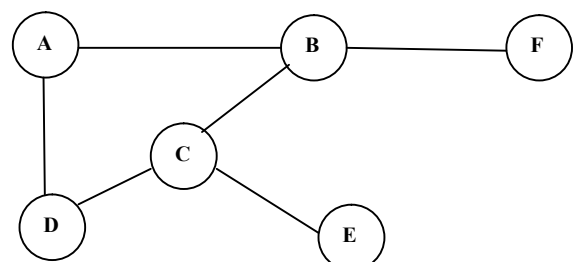
- (iv) What would be the Postfix notation for the given equation?
 $a+(b*c(d/e^f)*g)*h$
(a) $ab*cdef/^*g-h+$ (b) $abcdef^/*g*h*+$
(c) $abcd*^ed/g*-h*+$ (d) $abc*de^fg/*-*h+$
- (v) A variant of the linked list in which none of the node contains NULL pointer is?
(a) Singly Linked List (b) Doubly Linked List
(c) Circular Linked List (d) Reversed Linked List.
- (vi) The searching technique that takes $O(1)$ time to find a data is
(a) Linear Search (b) Binary Search (c) Hashing (d) Tree Search.
- (vii) Given a 5×3 matrix where 10 elements are zeroes, the order of its corresponding triplet array is
(a) 3×5 (b) 6×3 (c) 3×10 (d) 11×3
- (viii) Which of the following are true for a B-tree of order 5?
(a) The root can have a maximum of 5 child nodes
(b) Each leaf node has at least 2 key stored in it
(c) The heights of leaf nodes may differ
(d) None of the above is true.
- (ix) Complete the following recursive implementation to find the sum of digits of number:

```
int recursive_sum_of_digits(int n)
{
    if(n==0)
        return 0;
    else
        return _____;
}
int main()
{
    int n = 123;
    int ans = recursive_sum_of_digits(n);
    printf("Result = %d", ans);
    return 0;
}
```


(a) $(n / 10) + \text{recursive_sum_of_digits}(n \% 10)$
(b) $(n \% 10) + \text{recursive_sum_of_digits}(n / 10)$
(c) $(n) + \text{recursive_sum_of_digits}(n \% 10)$
(d) $(n \% 10) + \text{recursive_sum_of_digits}(n \% 10)$.

- (x) In the given graph identify the cut vertices

- (a) A and D (b) D and E
(c) C and B (d) F and C.



Group – B

2. (a) The running time of an algorithm, $T(n)$, where 'n' is the input size, is given by $T(n) = 1$, for $n = 1$ and $T(n) = T(n/3) + c$, for $n > 1$. Find the order of the algorithm.
- (b) Write a function to remove duplicates from an ordered array, without using any second array.
For example, if the original content of the array is: a, a, c, d, q, q, r, s, u, w, w, w, w; then the final content of the array should be a, c, d, q, r, s, u, w.
- (c) Write one difference between linear and non-linear data structure.
- 4 + 7 + 1 = 12**
3. (a) Say we have a 4x4 matrix with the following values,
0 2 0 0
1 0 0 3
0 0 5 0
0 4 0 0
What will be the triplet format of the above mentioned matrix? Check whether transferring the above mentioned matrix to triplet format is at all beneficial or not?
- (b) Write a function which will take any number n as its argument. The function will break this number into its individual digits and then store every single digit in a separate node thereby forming a linked list. The function must return the head node address of the created linked list at the end. (for example, if the number is 13579, then there will be 5 nodes in the list containing nodes with values 1, 3, 5, 7, 9).
- (c) Find the Big-Oh for $T(n) = 48n^{100} + 2^{n+2} + 3n^2 + 100$.
- (3 + 2) + 5 + 2 = 12**

Group – C

4. (a) Evaluate the result of the given infix expression using proper data structure:
 $2*(5*(3+6))/15 - 2$.
Show every step clearly.
- (b) Discuss the key features of tail recursion. Explain with suitable example how it is different from a normal recursive function.
- (c) Is deque a FIFO data structure? Explain.
- 5 + 4 + 3 = 12**
5. (a) Write a small C function (no need for main, header files etc.) to print the moves of the well-known Tower of Hanoi problem using recursion. Assume that the smallest disc is numbered as 'disc 1' and the largest is 'disc n'. Each of the three pegs is represented by a char (datatype) – source ('S'), destination ('D') and temporary ('T'). Given below are some examples of the moves that you will print –
Move the disc # 12 from T to D

Move the disc # 4 from S to T

- (b) Now just write the output of your function if it is called with $n = 4$, where n is the number of discs.
- (c) What will the following function return if it is called from the main as `XXXX(5, 0, 1)`? Can you identify what the following function is calculating? It is something well-known.

```
int XXXX(int n, int a, int b)
{
    if (n == 0)
        return a;
    if (n == 1)
        return b;
    return XXXX(n - 1, b, a + b);
}
```

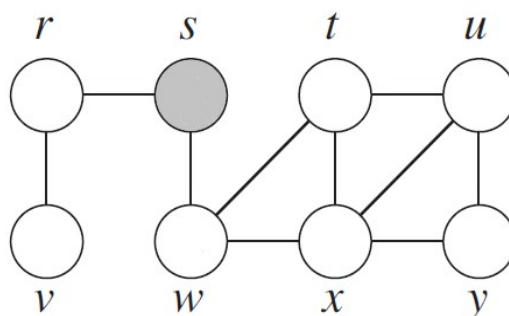
$$5 + 2 + (4 + 1) = 12$$

Group - D

6. (a) Reconstruct a binary tree from its traversal sequences given below:
Inorder : J E N K O P B F A C L G M D H I
Postorder : J N O P K E F B C L M G H I D A
What will be the preorder traversal sequence of the tree?
- (b) Draw the different stages of an initially-empty AVL tree by inserting nodes in the following sequence: 7, 10, 12, 37, 32, 22, 25, 27, 35, 45, 52. Also show the balance factors of each node.

$$(5 + 2) + 5 = 12$$

7. (a) What are the two main ways to represent a graph? Write the space required to store each of those. When do you choose which representation?
- (b) Robin has drawn the BFS tree of a graph G with 100 nodes. It contains 90 edges. Is this possible? Did he draw it correctly? Justify your answer.
- (c) Suppose you are performing a depth-first traversal on the graph given, with s as the starting node. Write just the discovery time and finishing time of each vertex and draw the depth-first tree. Assume that in each of the adjacency lists the vertices are in alphabetical order, and so understand that now the answers will be unique.



$$(2 + 2 + 1) + 2 + 5 = 12$$

Group – E

8. (a) Construct a max-heap using the given sequence of elements, 2, 8, 6, 1, 10, 15, 3, 12, 11.
- (b) Derive the worst-case time complexity of a binary search.
- (c) Using linear probing method to insert the following elements in a table of size 7: 76, 93, 40, 47, 10, 55
Use the hash function $(h(k) + i) \bmod 7$, $h(k) = k \bmod 7$, $i = 0.6$.

$$3 + 4 + 5 = 12$$

9. (a) Fill up the following table with the appropriate asymptotic complexities:-

Type of Sorting	Best Case	Average Case	Worst Case
Naïve Bubble Sort	$O(n^2)$		
Improved Bubble Sort		$O(n^2)$	
Insertion Sort			
Selection Sort			
Counting Sort		$O(n+k)$	

- (b) Write an algorithm to implement the Interpolation Search. Under what condition does the Interpolation Search behave like sequential search?
- (c) What is the time complexity of the worst case of linear search?

$$6 + (4 + 1) + 1 = 12$$

Department & Section	Submission Link
CSE A	https://classroom.google.com/c/MjYzNzg3MDAw/a/Mjg4NDMzMzM3MjA5/details
CSE B	https://classroom.google.com/c/MTE4NzY4MjM4NjM4/a/Mjg4NDI0ODI0NzY5/details
CSE C	https://classroom.google.com/u/0/w/MTMwMDU2NDk2NDI1/tc/Mjk0MzczNjYwMTc4
LATERAL	https://classroom.google.com/c/MjUzMzUzMdG1MzI1/a/Mjg4NDI3Mzg4NTQy/details



DS and Algo (CSEN 2101 Solution) - 2020

- 1 i) b ii) d iii) e iv) b v) e
 vi) c vii) b ~~viii) d~~ viii) d ix) b x) c

2 a. $T(n) = 1$, for $n=1$ — (1)

$T(n) = T(n/3) + c$, for $n > 1$ — (2)

$$T(n) = T\left(\frac{n}{3}\right) + c$$

$$= T\left(\frac{n}{3^1}\right) + 1 \cdot c$$

Substituting eqn. (2)

$$= T\left(\frac{n}{3^2}\right) + 2c$$

$$= T\left(\frac{n}{3^3}\right) + 3c$$

$$\vdots$$

$$= T\left(\frac{n}{3^k}\right) + kc \text{ — (3)}$$

∴ Terminating condition is $T(1) = 1$, to get $T(1)$, we should have $\frac{n}{3^k}$ as 1.

ie $\frac{n}{3^k} = 1 \Rightarrow 3^k = n$ Taking $\log_3$ on both sides

$$\log_3 3^k = \log_3 n \Rightarrow k \log_3 3 = \log_3 n$$

Now, $\log_3 3 = 1 \therefore k = \log_3 n$ — (4)

Substituting (4) in (3)

$$T(n) = T(1) + c \cdot \log_3 n = c \cdot \log_3 n \therefore T(n) = O(\log_3 n)$$

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initial array size = n.

b) for ($i=0$; $i < n-1$; $i++$)

if ($arr[i] \neq arr[i+1]$)

$arr[j++] = arr[i];$

$arr[j++] = arr[n-1];$

final array size = j

c) In linear DS elements are sequentially connected and each element is traversible through a single run. In non-linear DS elements are hierarchically connected and are present at various levels.

3(a)

~~4 4 5~~

~~1 1 2~~

4 4 5

0 1 2

1 0 1

1 3 3

2 2 5

3 1 4

Original matrix: $4 \times 4 \times n$

Triplet matrix: $6 \times 3 \times n$.

original = beneficial.

(b) while ($n > 0$)

{

$k = n \% 10$; $n = n / 10$;

insert_at_beg(k);

}

c. The highest order term is ~~2~~ 2^{n+2} the Big-Oh for $T(n)$ will be $O(2^n)$. - 1 mark

If justification given, add another 1 mark.

4(a) append ')' to end of string and push '('

Input	Stack	O/P.	Rule
2	(	2	operand.
*	[*	2	operator
(	(*(	2	operator
5	(*(	25	operand.
*	(*(	25	op.
(	(*(	25	op.
3	(*(	253	operand.
+	(*(	253	operator
6	(*(	2536	operand
)	(*	2536+	'/' pop all op from stack until '(' is encountered.
)	(	2536+*	"
/	(/	2536+**	* & / have same precedence so popped.
15	(/	2536+**15	operand
-	(-	2536+**15/	/ has higher precedence than - hence popped.
2	(-	2536+**15/2	operand
)		2536+**15/2-	'/' encountered pop all until '(' found

- (b) Tail Recursion is a special case of recursion in which the last operation^{if any} performed by the function is either a recursive call or a value is returned w/o recursion (2)

Example: (2)

- (c) Dequeue is not a FIFO DS. as it allows insertion and deletion from both ends. - (1)

Explanation with example - (2)

↳ w/o example - (1)

(d)

```
5 (a) move(n, src, dest, temp)
{
    if (n == 1)
        printf("In Move %d from %c to %c", n, src, dest);
    else
    {
        move(n-1, src, temp, dest);
        printf("In Move %d from %c to %c", n, src, dest);
        move(n-1, temp, dest, src);
    }
}
```

$$(b) \quad 1 \rightarrow S \rightarrow T$$

$$2 \rightarrow S \rightarrow D$$

$$1 \rightarrow T \rightarrow D$$

$$3 \rightarrow S \rightarrow T$$

$$1 \rightarrow D \rightarrow S$$

$$2 \rightarrow D \rightarrow T$$

$$1 \rightarrow S \rightarrow T$$

$$4 \rightarrow S \rightarrow D$$

$$1 \rightarrow T \rightarrow D$$

$$2 \rightarrow T \rightarrow S$$

$$1 \rightarrow D \rightarrow S$$

$$3 \rightarrow T \rightarrow D$$

$$1 \rightarrow S \rightarrow T$$

$$2 \rightarrow S \rightarrow D$$

$$1 \rightarrow T \rightarrow D$$

c

$$0 - 0$$

$$1 - 1$$

$$2 \rightarrow 1$$

$$3 \rightarrow 2$$

$$4 \rightarrow 3$$

$$5 \rightarrow 5$$

n^{th} term of fibonacci series.

return 52.

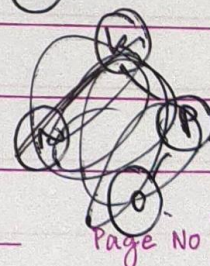
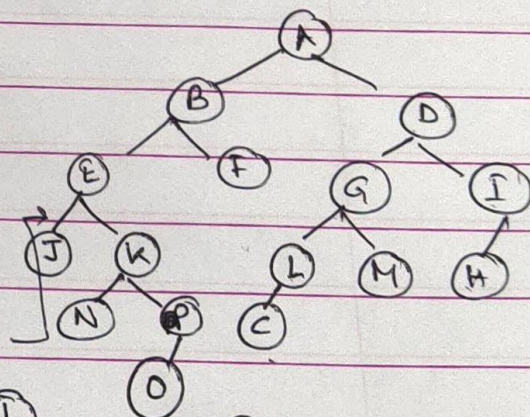
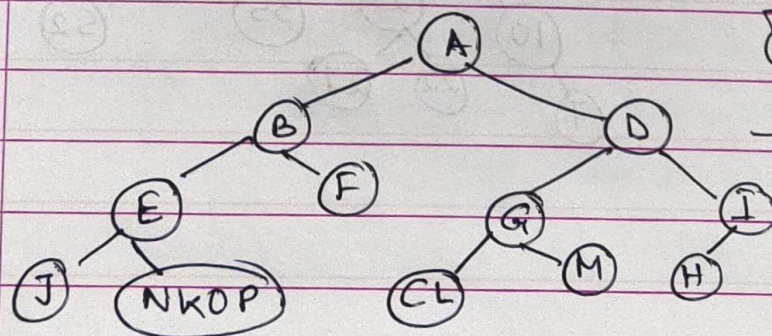
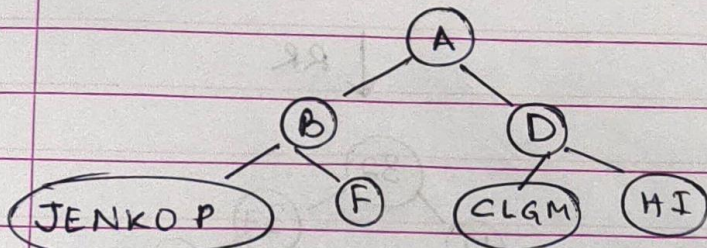
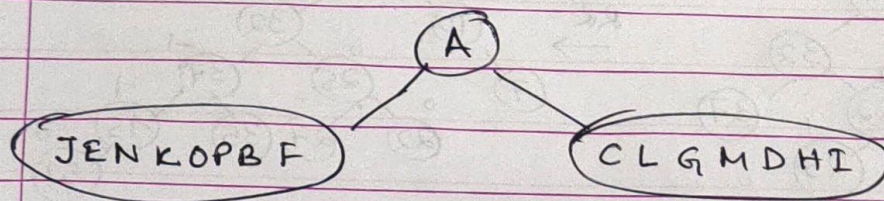
1	3	5
2	2	3
3	1	2
4	1	1
5	0	1

6a)

In order — L R_T R.

Post order — L R R_T.

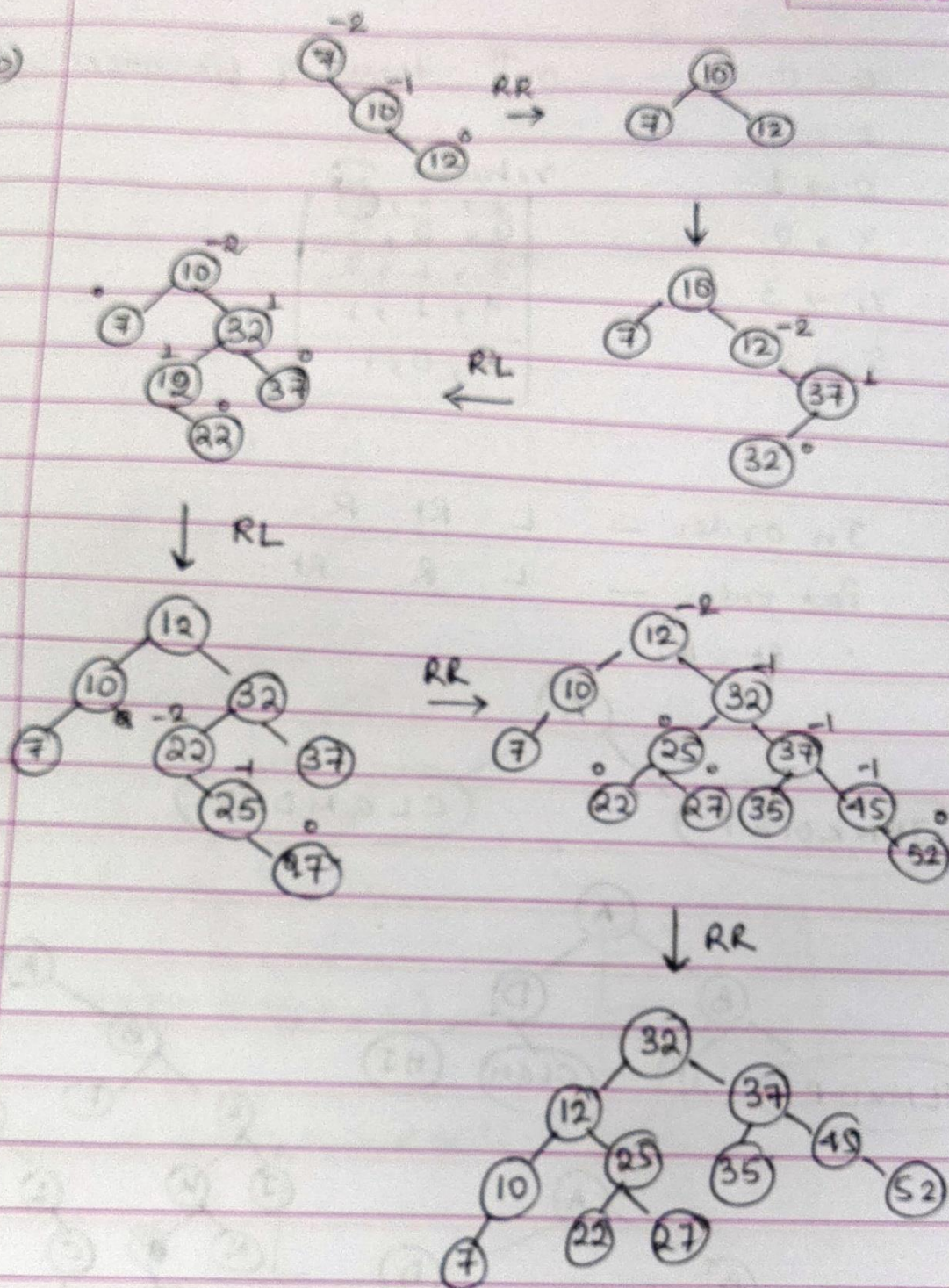
∴ R_T = A.



Pre order: A B E J K N P O F D G L C M I H

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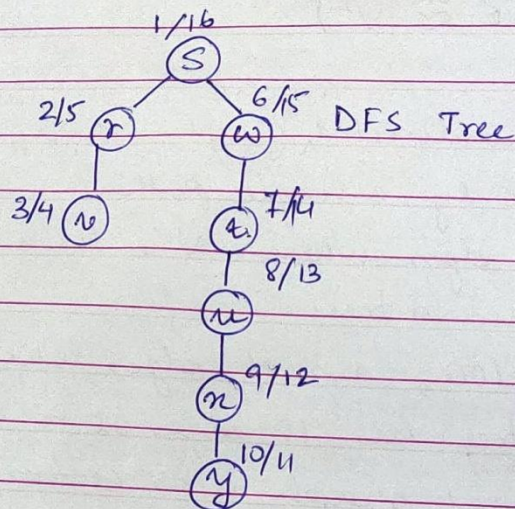
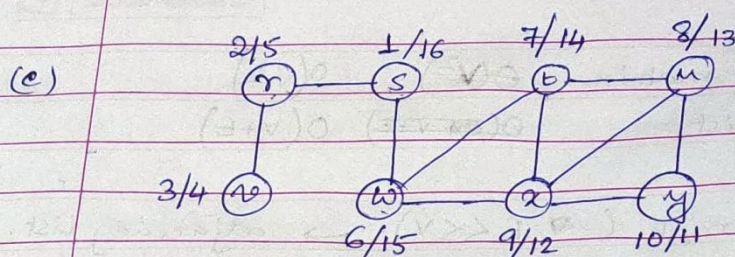
(b)

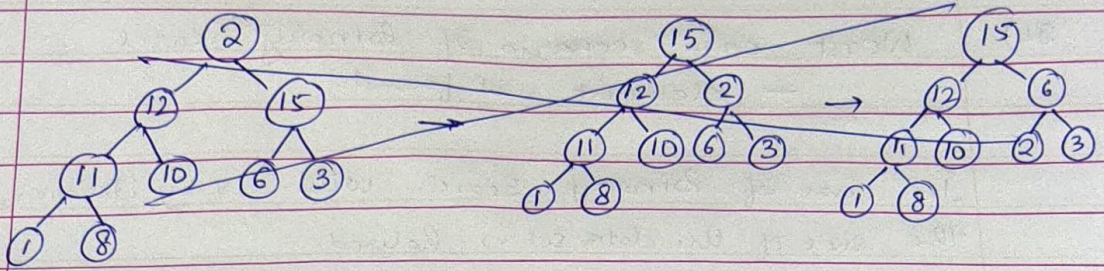


7a) Adjacency matrix $O(V^2)$
 Adjacency list. $O(V+E)$

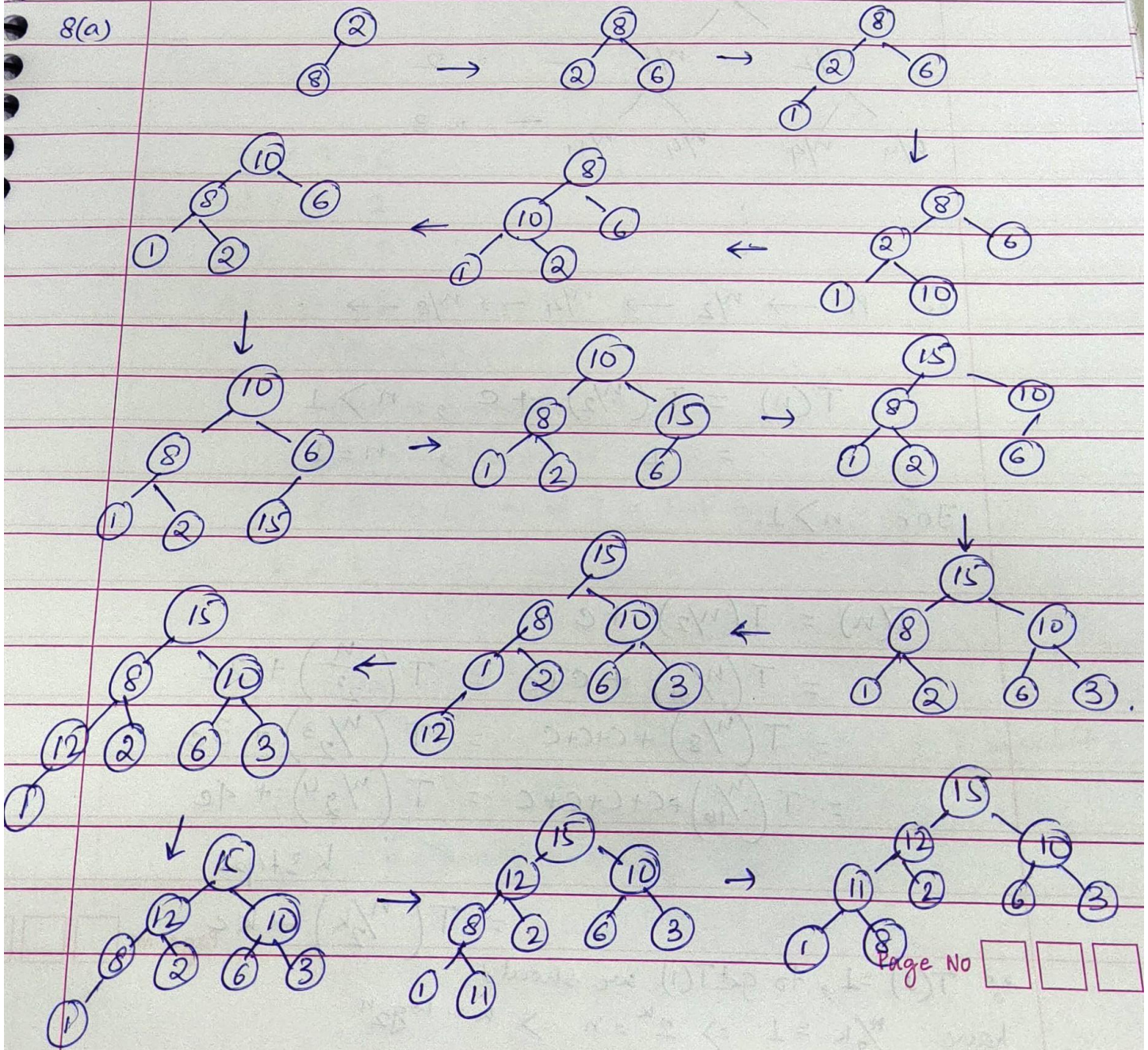
Sparse graph ($E \ll V$) $\rightarrow$ adjacency list.
 Dense graph ($E \approx V(V-1)/2$ OR $E \approx V^2$) $\rightarrow$ matrix/List.

b) A tree is an undirected ^{acyclic} graph where ^{any} 2 vertices are connected by only a single path.
 Hence has $N-1$ edges where N is the no. of vertices.
 BFS tree is also a tree.
 Hence, if $N=100$, no. of edges should be $=99$
 $\therefore$ Robin's tree has 90 edges it is not possible.
 if the original graph was ~~conn~~ a connected graph.
 BFS does not work for disconnected graph.



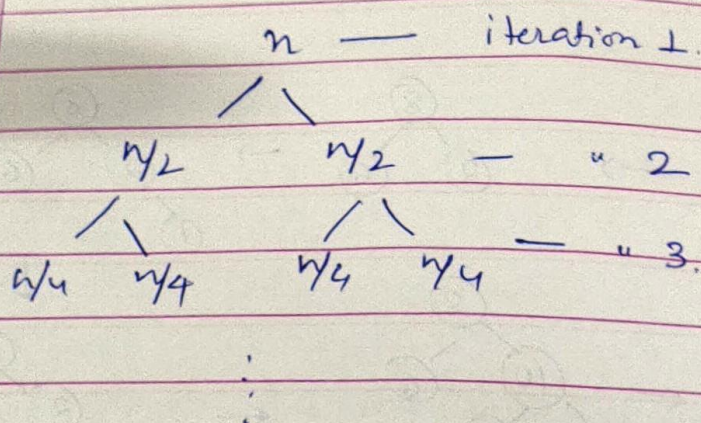


8(a)



8b) Worst case scenario of Binary Search
— element not found.

In case of Binary Search with every iteration, the size of the data set is halved.



$$\therefore n \rightarrow n/2 \rightarrow n/4 \rightarrow n/8 \rightarrow \dots$$

$$T(n) = T(n/2) + c, \quad n > 1$$

$$= 1, \quad n = 1$$

For $n > 1$.

$$T(n) = T(n/2) + c$$

$$= T(n/4) + c + c = T\left(\frac{n}{2^2}\right) + 2c$$

$$= T(n/8) + c + c + c = T\left(\frac{n}{2^3}\right) + 3c$$

$$= T\left(\frac{n}{16}\right) + c + c + c + c = T\left(\frac{n}{2^4}\right) + 4c$$

⋮
k steps

$$= T\left(\frac{n}{2^k}\right) + kc$$

$\therefore T(1) = 1$, to get $T(1)$ we should

have $\frac{n}{2^k} = 1 \Rightarrow 2^k = n \Rightarrow k = \log_2 n$

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$$\begin{aligned} \therefore T(n) &= T(1) + c \cdot \log_2 n \\ &= 1 + c \cdot \log_2 n \end{aligned}$$

$$\therefore T(n) \text{ is in } O(\log_2 n)$$

c.

0	1	2	3	4	5	6
47	55	93	10		40	76

$$(h(k) + i) \bmod 7$$

$$76 \div 7 = 6$$

$$93 \div 7 = 2$$

$$40 \div 7 = 5$$

$$47 \div 7 = 5 \text{ [Collision]}$$

$$(5 + 0.6) \bmod 7 = 5.6$$

$$(5.6 + 0.6) \bmod 7 = 6.2$$

$$(6.2 + 0.6) \bmod 7 = 6.8$$

$$(6.8 + 0.6) \bmod 7 = 7.4 \bmod 7 = 0.4 \approx 0$$

$$10 \div 7 = 3$$

$$55 \div 7 = 6$$

$$(6 + 0.6) \bmod 7 = 6.6$$

$$(6.6 + 0.6) \bmod 7 = (7.2) \bmod 7 = 0.2 \text{ mod } 7$$

$$(0.2 + 0.6) \bmod 7 = 0.8 \bmod 7$$

$$(0.8 + 0.6) \bmod 7 = 1.4 \bmod 7 \approx 1$$

Q(a)		Best	Avg	Worst.
Native	B. Sort	$O(n^2)$	$O(n^2)$	$O(n^2)$
Improved	B. Sort	$O(n^2)$	$O(n^2)$	$O(n^2)$
Insertion	Sort	$O(n)$	$O(n^2)$	$O(n^2)$
Selection	Sort	$O(n^2)$	$O(n^2)$	$O(n^2)$
Counting	Sort	$O(n+k)$	$O(n+k)$	$O(n+k)$
		Technically Ω	Θ	

(b) If the data is not uniformly distributed Interpolation Search behaves as sequential search

```

isearch (int arr[], int lo, int hi, int x)
{
    int pos;
if (lo < hi)
    if (lo ≤ hi && x ≥ arr[lo] && x ≤ arr[hi])
    {
        pos = lo +  $\frac{(hi - lo)}{arr[hi] - arr[lo]} * (x - arr[lo])$ 

        if (arr[pos] == x)
            return x;
        elseif (arr[pos] < x)
            return isearch(arr, pos+1, hi, x);
        else
            return isearch(arr, lo, pos-1, x);
    }
    return -1;
}

```

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